

10

Lymphatic and immune systems

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The **lymphatic and immune systems** are formed by **lymphatic vessels, organs and immune cells** with the substances secrete. Lymphatic vessels are responsible for **unidirectional drainage of extracellular fluid into the bloodstream**. They originate as blind-ended tubes in intercellular spaces (lymphatic capillaries), which drain into the larger lymphatic vessels before gathering into lymphatic trunks and flowing into the two main lymphatic ducts. Lymphatic fluid (or simply “lymph”) is drained at the venous angle into the venous systemic blood. The lymphatic vessels are located in almost every tissue **except for cartilage, cornea, bone marrow, placenta and the central nervous system**. In these sites drainage of extracellular fluid occurs in a different way. The lymphatic system is integrally related to the immune system. They cooperate together in forming immune reactions that protect the body from infections, cancerous growths and toxic substances.

Function

Lymphatic system

1. **immune function:** cellular and humoral immunity (antibody-mediated immune system)
2. **extracellular fluid drainage** into the systemic blood circulation
3. **transport of proteins, chylomicrons and fatty acids** into the systemic blood circulation
4. **elimination of senescent erythrocytes** by macrophages in the spleen (90 %), bone marrow and liver

Immune system

Primary immune organs

1. **de novo genesis of immune cells**
the site of the process of differentiation and maturation
2. **the bone marrow is responsible for the production of all immune cells**
(granulocytes, monocytes and lymphocytes)

Secondary immune organs

1. responsible for **interaction between naive lymphocytes and antigen-presenting cell (APC)**
2. initiation of **specific immune responses** and **genesis** of immune memory cells
3. responsible for **the production of effective lymphocytes against specific agents**

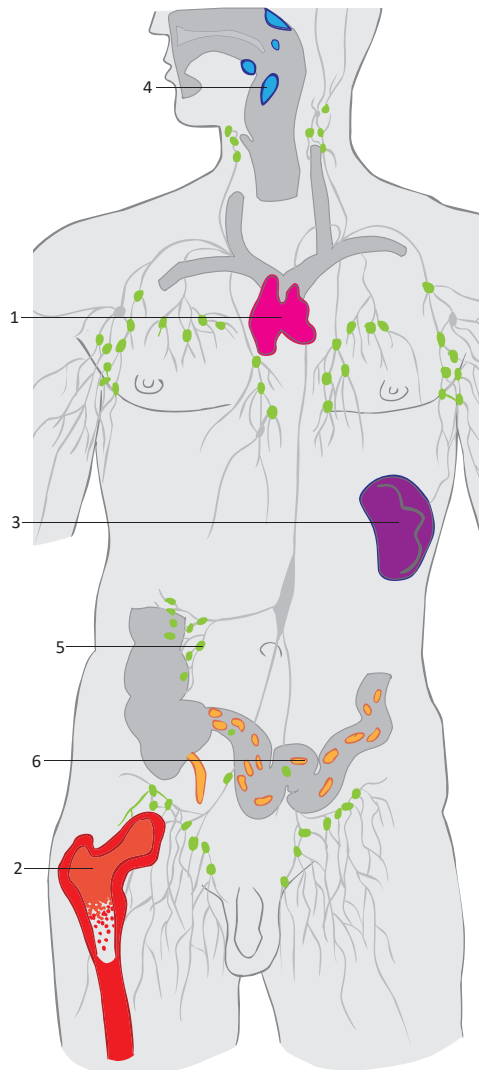
Organs

Primary (central) organs

- 1 **Thymus**
- 2 **Bone marrow** (*medulla ossium*)

Secondary (peripheral) organs

- 3 **Spleen** (*splen/lien*)
- 4 **Tonsils** (*tonsillae*)
- 5 **Lymph nodes** (*nodi lymphoidei*)
- 6 **Lymphoid nodules** (*noduli lymphoidei*)



Daily production of lymph is about 60 ml/kg.

Lymph (*lymph*) is formed from interstitial fluid that flows into lymphatic capillaries. It is fundamentally different and easily distinguishable from plasma by its relative lack of protein.

Chylus is intestinal lymph. It has a milky appearance and contains chylomicrons.

Chylomicrons are products of enterocytes. They transport triacylglycerol and cholesterol from the intestinal lumen through lymphatic vessels into the bloodstream.

The central nervous system does not contain any lymphatic vessels. Instead, interstitial fluid drains to the outside of the brain alongside the pulsating cerebral arteries, through the cribriform plate of the ethmoidal bones and also partially into the CSF.

Embryonic development of the lymphoid organs:

The thymus is derived from the endoderm of the ventral part of the third pharyngeal pouch. Cells from the neural crest also contribute to its development.

The spleen develops by an accumulation of mesenchyme cells within the dorsal mesogastrium.

The tonsils arise from the endoderm of the second pharyngeal pouch. In the 3rd month of intrauterine life the epithelium forms a reticular framework. This is followed by migration of lymphocytes, which form lymphoid nodules.

Lymph nodes develop either from collapsing lymphatic sacs during the third month (primary lymph nodes) or from aggregations of mesenchymal cells that proliferate along capillary plexuses (secondary lymph nodes).

Clinical notes

Lymphography (lymphangiography) is a medical imaging technique in which lymphatic vessels and lymph nodes are visualised injecting contrast dye and taking X-ray pictures. **Lymphoscintigraphy** is a similar technique performed by injecting radioisotopes and imaging with positron emission tomography (PET) scans.

Lymphoedema is swelling due to congestion of lymphatic fluid. It occurs when lymph drainage is insufficient and may progress to elephantiasis, a condition where parts of the skin, subcutaneous tissues or entire regions (scrotum, limbs) swell up to large or even massive proportions.

Lymphatic capillaries are blind-ended tubes with a special microscopic structure that ensures a **unidirectional flow of extracellular fluid**. Lymphatic capillaries drain into the bigger lymphatic vessels. **Lymph nodes are found along the network of lymphatic vessels**. Lymphatic vessels contain valves, which supports the unidirectional fluid to the lymphatic trunks and ducts. **The lymphatic ducts drain lymph into the venous blood at the venous angle**.

1 Lymphatic capillaries (*vasa lymphocapillaria*)

- collect extracellular fluid with metabolic substances, antigens (viruses, bacteria and/or pre-tumor or tumor cells) and antigen-presenting cells
- form a lymphocapillary network (*rete lymphocapillare*), whose wall is made of only one layer of endothelium (running together with blood capillaries and venules; in the small intestine, run in the axis of a villus)

2 Lymphatic vessels (*vasa lymphoidea*)

- are thin-walled vessels containing valves
- their course is interrupted by lymph nodes
- in organs, there are subcapsular and deep lymphatic networks
- in extremities, there are superficial and deep collectors accompanying particular veins
 - the superficial collectors are located in subcutaneous tissue and empty into the cubital and axillary nodes in the upper limb, and the popliteal and inguinal nodes in the lower limb
- the deep collectors run together with the deep vessels

2.1 Collectors of the upper extremity

- 2.1.1 **Lateral collectors** – to the axillary lymph nodes along the cephalic vein
- 2.1.2 **Medial collectors** – to the axillary lymph nodes along the basilic vein (drain the superficial cubital lymph nodes of the cubital region)
- 2.1.3 **Anterior collectors** – to the superficial cubital lymph nodes along the median antebrachial vein

2.2 Collectors of the lower extremity

- 2.2.1 **Medial collectors** (10–15) – to the superficial inguinal lymph nodes, along the great saphenous vein
- 2.2.2 **Lateral collectors** (1–3) – to the superficial inguinal lymph nodes, along the tributaries of the great saphenous vein
- 2.2.3 **Posterior collectors** (2–3) – to the superficial popliteal lymph nodes, along the small saphenous vein

- 2.3 **Deep lymphatic vessels** – accompany the deep vessels of the lower extremity,
 - to the deep popliteal lymph nodes and the deep inguinal lymph nodes

3 Lymphatic trunks (*trunci lymphatici*)

Paired

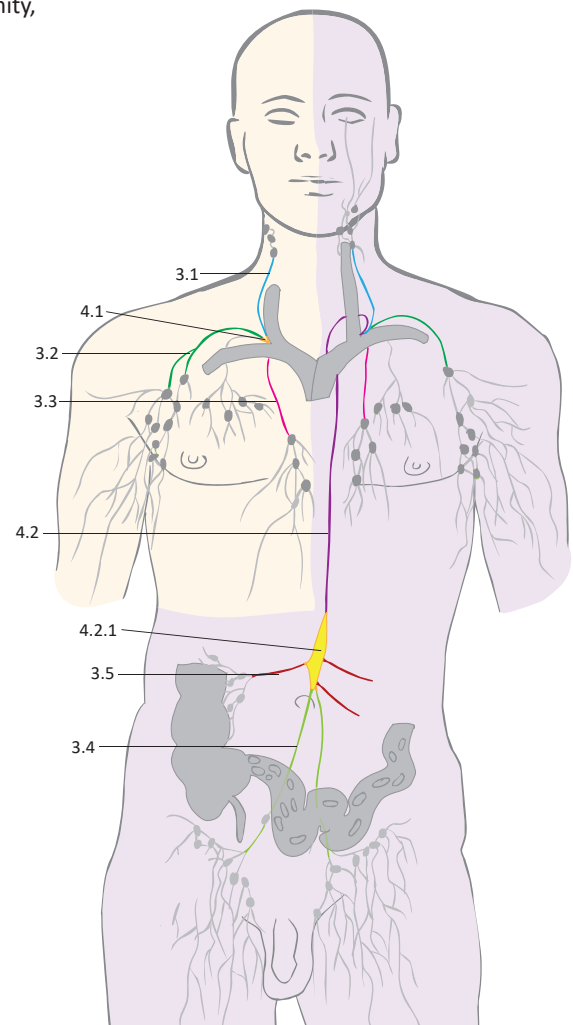
- 3.1 **Jugular trunk** (*truncus jugularis*)
 - lymphatic drainage from half of the head and neck
- 3.2 **Subclavian trunk** (*truncus subclavius*) – lymphatic drainage from the collectors of the upper extremity
 - the axillary lymphatic plexus is a network of lymphatic vessels and lymph nodes in the axilla, which drains in the subclavian trunk
- 3.3 **Bronchomediastinal trunk** (*truncus bronchomediastinalis*)
 - lymphatic drainage from half of the thorax
- 3.4 **Lumbar trunk** (*truncus lumbalis*) – lymphatic drainage from the lower extremity and half of the pelvis

Unpaired

- 3.5 **Intestinal trunk** (*truncus intestinalis*)
 - lymphatic drainage from the unpaired abdominal organs

4 Lymphatic ducts (*ductus lymphatici*) – irregular drainage areas

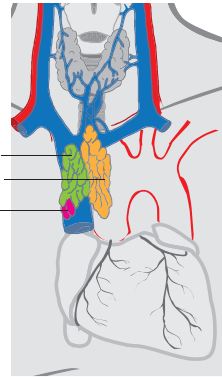
- 4.1 **Right lymphatic duct** (*ductus lymphaticus dexter*)
 - a short duct originating from the 3 main lymphatic trunks: the right jugular, right subclavian and right bronchomediastinal trunks
 - collects lymph from the right half of the head and neck, right upper extremity and right half of the thoracic cavity
 - there is an exception: lymphatic drainage from the right half of the heart goes to the thoracic duct
- 4.2 **Thoracic duct** (*ductus thoracicus*)
 - originates from the lumbar trunks
 - collects lymph from both lower extremities, pelvis, abdominal cavity, the left half of the thoracic cavity, left upper extremity and the left half of the head and neck
 - is divided into 4 parts according to its course: lumbar, abdominal, thoracic and cervical parts
 - 4.2.1 **Cisterna chyli** – a widened situated at the beginning of the thoracic duct at the level of T11–L1
 - there is an exception: lymphatic drainage from the 4th to 10th segment of the left lung and left half of the heart runs to the right lymphatic duct



The thymus is a lymphoepithelial organ located in the **ventral part of the superior mediastinum** just behind the sternum. It contains T-lymphocytes undergoing maturation. The T-lymphocytes are selected in the thymus to recognise self from non-self, meaning they are programmed to recognise and attack foreign antigens that may be a potential threat to the organism. **The thymus is the largest and most active during the neonatal period. During the teenage years it begins to atrophy and involute.** After the age of around fifty, the thymus is almost completely replaced by an adipose tissue. However, residual thymus tissue may persist throughout the rest of life.

Structure

- 1 **Right lobe** (*lobus dexter*)
- 2 **Left lobe** (*lobus sinister*)
- 3 **Lobules** (*lobuli*)
- 4 **Cortical septa** (*septa corticalia*) – bands of mediastinal fibrous tissue in the cortex
- 5 **Cortex** – composed mainly of T-lymphocytes
- 6 **Medulla** – contains epithelial reticular tissue, fewer T-lymphocytes and corpuscles of Hassal

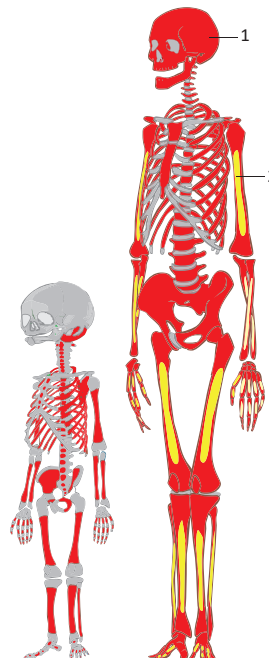


Function

1. **Programming of T cells progenitors**
2. **central tolerance** – distinction between self and non-self antigens
3. **most of the T-lymphocytes are eliminated** by apoptosis (programmed cell death) during the process of positive and negative selection
4. **selected T-lymphocytes migrate to the secondary lymphatic organs**
5. **endocrine function:** influences T-lymphocyte proliferation and maturation, regulates their functions and influences the migration of macrophages

From the eight month of intrauterine development, **bone marrow is the sole organ of haematopoiesis**. It is situated **inside the medullary cavity in long bones and between the trabeculae of cancellous bone**. Throughout life, bone marrow slowly transforms from active hematopoietic red marrow into inactive adipose yellow marrow .

- 1 **Red marrow** (*medulla osseum rubra*)
 - **the main haematopoietic organ**
 - contains stem cells, which differentiate into leukocytes, erythrocytes and megakaryocytes
 - platelets (thrombocytes) are formed by disintegration of megakaryocytes
 - **the reticular framework of bone marrow** is composed of reticular connective tissue, sinusoids formed by endothelial cells and macrophages
 - **at the age of 20 red marrow is replaced by yellow marrow** in the medullary cavity of all long bones except for the proximal ends of the humerus and femur
 - **red marrow persists in cancellous bone** in the epiphyses of long bones, short bones and flat bones (pelvis, sternum, ribs and in the diploe of certain regions of the skull)
- 2 **Yellow marrow** (*medulla osseum flava*)
 - is composed of reticular connective tissue with a high content of adipocytes
 - has the ability to reactivate into red marrow



The weight of the thymus is age related. In newborns it is 16 g (10–35 g), in adults 20–50 g and among the elderly 5–15 g.

Hassall's corpuscles / thymic corpuscles are structures in the medulla of the thymus. Their function is still not fully known.

The size of the spleen is 10–13 cm in length, 6–8 cm in width and 4 cm in thickness.

The weight of the spleen is 140–160 g in men and 120–150 g in women. A weight of even 200 g can still be considered physiological because the weight is highly dependent on the extent of blood flow.

Clinical notes

A sternal puncture is a diagnostic procedure in which a needle is inserted into the bone marrow of the manubrium of the sternum to collect a sample for cytological evaluation in blood disorders and malignancies.

A trephine biopsy is a procedure in which a cylindrically shaped core of bone is obtained for histopathological evaluation. Unlike sternal puncture, this procedure shows the structure and cellularity of the bone marrow including its supportive connective tissue, which can help to diagnose on-going disorders. In children up to the age of two, bone marrow from the tibia is collected. In the elderly, samples are drawn from the hip bone, usually from the area of the posterior superior iliac spine and occasionally from the anterior superior iliac spine.

Splenomegaly is the pathological enlargement of the spleen. It has a wide variety of causes such as viral mononucleosis, liver disease, lymphoma and leukaemia.

2-stage rupture of the spleen is a life threatening situation and a diagnostic challenge in closed abdominal trauma. After injury to the pulp, the splenic parenchyma ruptures and blood accumulates under the capsule. The pressure under the capsule rises and eventually causes the capsule to rupture, which causes blood to spill out into the peritoneal cavity.

Splenectomy is the surgical removal of the spleen. Without the spleen the human body is susceptible to infections caused by encapsulated bacteria such as pneumococcus, meningococcus and haemophilus. Thus vaccination against these pathogens is paramount.

The spleen is the largest secondary lymphoid organ. **It is active in the initiation of immune reactions against antigens filtered from the blood stream and participates in the degradation of blood cells.** It is an **intraperitoneal organ**, located in the left hypochondrium in the area between the 9th and 11th rib. The pleural cavity extends behind the diaphragmatic surface of the spleen. In a healthy human, the spleen is not palpable during physical examination.

External structure

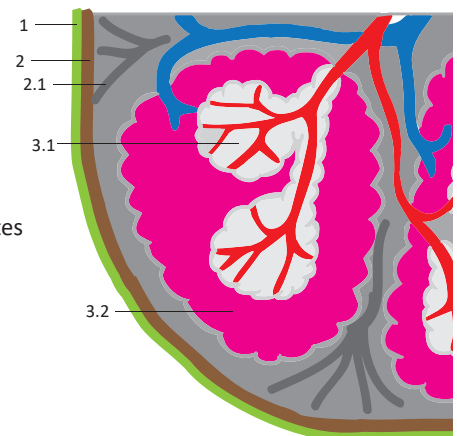
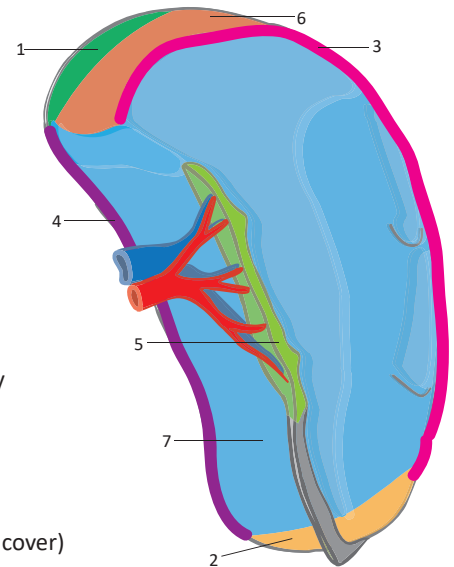
- 1 **Posterior extremity** (*extremitas posterior*) – the dorsal end of the spleen
- 2 **Anterior extremity** (*extremitas anterior*) – the ventral end of the spleen
- 3 **Superior border** (*margo superior*) – the upper front sharp margin
- 4 **Inferior border** (*margo inferior*) – the lower back more obtuse margin
- 5 **Splenic hilum** (*hilum splenicum*) – the entry point for the splenic vessels and nerves
- 6 **Diaphragmatic surface** (*facies diaphragmatica*) – the dorsocranial surface adjacent to the diaphragm
- 7 **Visceral surface** (*facies visceralis*) – the ventral surface in contact with the stomach, left kidney, pancreas and left colic flexure (splenic flexure)
- 8 **Splenic recess of omental bursa** (*recessus splenicus bursae omentalis*)

Supporting ligaments

- 1 **Splenorenal/lienorenal ligament** (*ligamentum splenorenale*) – between the spleen and kidney
- 2 **Gastrosplenic ligament** (*ligamentum gastrosplenicum*) – between the spleen and stomach
- 3 **Splenicocolic ligament** (*ligamentum splenicolicum*) – between the spleen and transverse colon

Internal structure

- 1 **Serosa / serous coat** (*tunica serosa / peritoneum viscerale*) – the peritoneum (a thin outer cover)
- 2 **Capsule** (*tunica fibrosa*) – fibrous capsule (thick inner cover)
 - 2.1 **Splenic trabeculae** (*trabeculae splenicae*) – fibrous septa extending from the capsule into the parenchyma
- 3 **Splenic pulp** (*pulpa splenica*) – the main substance of the spleen
 - 3.1 **White pulp** (*pulpa alba*)
 - **Periarteriolar lymphoid sheath** (*vagina lymphoidea periarteriolaris* – PALS) – populated largely by T-lymphocytes
 - **Marginal zone** (*zona marginalis*) – a region composed of macrophages (antigen-presenting cell) and B-lymphocyte
 - **Splenic lymphoid nodules** (*noduli lymphoidei splenici*) – contain many lymphocytes
 - 3.2 **Red pulp** (*pulpa rubra*) – is composed of connective tissue – constitutes the splenic cords of Billroth and the splenic sinuses between them



Cross section of the spleen

Functions

- elimination of microorganisms from the blood
- specific immune reaction against antigens filtered from blood
- activation and proliferation of lymphocytes and synthesis of antigens and antibodies
- degradation of erythrocytes and thrombocytes
- erythropoiesis until the 7th month of intrauterine life

Blood supply

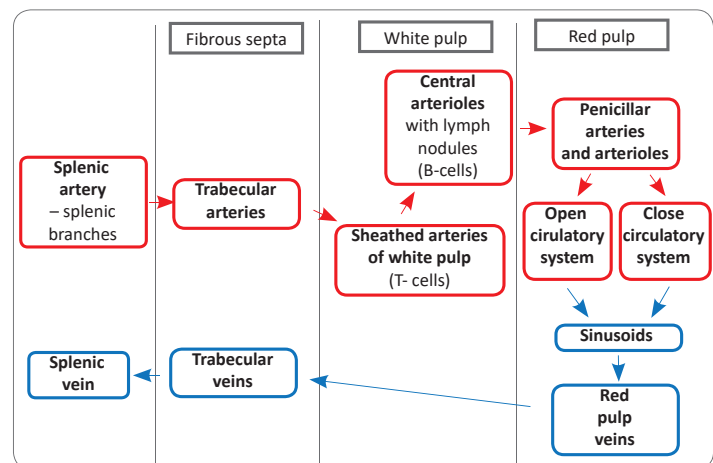
Arterial supply:

Splenic artery (*arteria splenica*), which branches into:

- 1. **Splenic branches** (*rami splenici*)
- 2. **Trabecular arteries** (*arteriae trabeculares*)
- 3. **Sheathed arteries of white pulp** (*arteriolae vaginatae pulpae albae*) – within the periarteriolar lymphoid sheath
- 4. **Central (nodular) arterioles** (*arteriolae centrales/nodulares*)
- 5. **Penicillar arteries** (*arteriolae penicillares*)
- 6. **Penicillar arterioles** (*arteriolae penicillares*)
- 7. **Open circulatory system** – blood is flows freely into the red pulp and then into the splenic sinuses
- 8. **Close circulatory system** – blood is kept within the splenic sinuses
- 9. **Sinusoids** (*vasa sinusoidea splenica*) – in the red pulp

Venous drainage:

- the **red pulp veins** (*venae pulpae rubrae*) into the **trabecular veins** (*venae trabeculares*)
- then to the **splenic vein** (*vena splenica*) and into the portal vein (*vena portae*)



Scheme of the vascular supply of the spleen

The tonsils form a **protective barrier** against foreign pathogenic particles at the **entrance to the respiratory and digestive tracts**. The tonsils are composed of small lymphoid nodules with germinal centres. They are located under the mucosa of the pharynx and root of the tongue and are arranged in a ring shape, called the **pharyngeal lymphoid ring / Waldeyer's ring** (*anulus lymphoideus pharyngis*). This ring is connected caudally to the lymphoid tissue of the larynx.

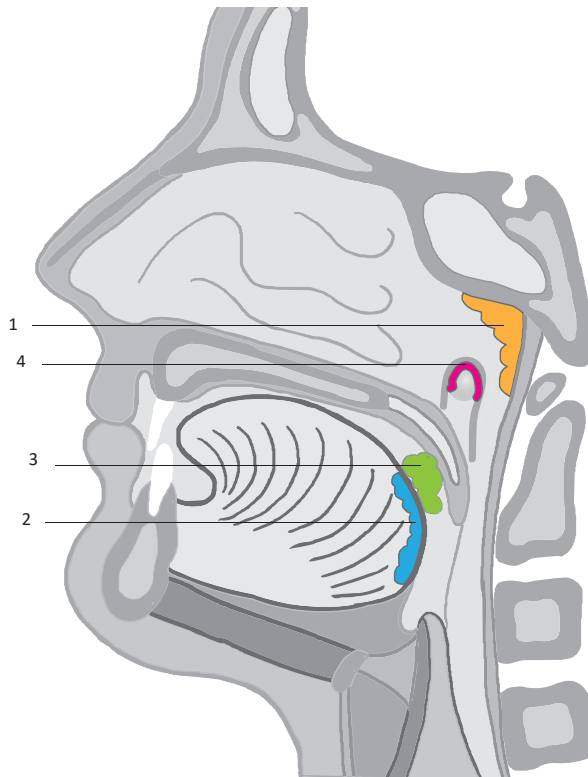
Pharyngeal lymphoid ring (*anulus lymphoideus pharyngis Waldeyeri*)

Unpaired tonsils

- 1 **Pharyngeal/nasopharyngeal tonsil / adenoid** (*tonsilla pharyngealis*)
 - **location:** the vault of the pharynx (*fornix pharyngis*)
 - **structure:** pits (*fossulae*), crypts (*cryptae*) and nodules (*noduli*)
 - is coated with a thin capsule
- 2 **Lingual tonsil** (*tonsilla lingualis*)
 - **location:** the root of the tongue (*radix linguae*)
 - **structure:** crypts (*crypta*) and nodules (*noduli*)

Geminate tonsils – bilateral tonsils

- 3 **Palatine tonsil** (*tonsilla palatina*)
 - **location:** the back of the throat in the tonsillar fossa (*fossa tonsillaris*) behind the palatoglossal arch (*arcus palatoglossus*) and in front of the palatopharyngeal arch (*arcus palatopharyngeus*)
 - **structure:** pits (*fossulae*), crypts (*cryptae*)
 - coated with a thick capsule
 - located within a large tonsillar bed
- 4 **Tubal tonsil** (*tonsilla tubaria*)
 - **location:** near the pharyngeal opening of the auditory tube (*ostium pharyngeum tubea auditivae*)
 - **structure:** crypts (*cryptae*)
 - the smallest tonsil



Sagittal section through the head and neck

Lymph nodes are small, self-contained organs surrounded by fibrous capsules and located along the course of lymphatic vessels.

Lymphoid nodules/follicles are masses of densely packed lymphocytes embedded in various tissues of the body including the cortex of lymph nodes, the spleen, the tonsils and the mucosa of the intestine. They are composed of a germinal centre consisting of various cells (dendritic cells, centrocytes and centroblasts) and a marginal zone, containing mainly immature B-cells. A primary nodule lacks the germinal centre. A secondary nodule has a germinal centre, which is formed after the first contact with an antigen.

Dendritic cells are a type of antigen presenting cell (APC).

Centroblasts and centrocytes are forms of activated B-cells, which gradually change into plasma cells or memory B-cells. Plasma cells produce antibodies. Memory cells are inactive cells that serve to provide a faster immune reaction on recurrent infection.

Clinical notes

Adenoid is a clinical term for the pharyngeal tonsil. Adenectomy is the surgical removal of the adenoid.

Lymphadenopathy is a term used for enlarged lymph nodes. Lymphadenopathy is usually painful in inflammation but usually painless in tumour metastasis.

Lateral bands is a clinical term for the dispersed lymphatic tissue in the lateral wall of the pharynx.

The sentinel node needs to be examined before surgical treatment of tumours. To perform a sentinel node biopsy, firstly a fluorescent or radioactive substance is injected into the tumour. The region is carefully dissected and the sentinel node is detected by a change in colour or radioactivity. The excised lymphatic tissue is then sent for a cytological examination for malignancy.

The TNM staging system

T (tumour) – the size of the primary tumour

N (nodes) – involvement of nearby (regional) lymph nodes

M (metastasis) – spread of cancer to distant body parts

Routes of metastatic spread

1. **transcoelomic spread** (spread and implantation of tumour cells on the peritoneum/pleura.)

2. **lymphogenous spread**

3. **haematogenous spread**

Nonencapsulated lymphoid tissue consists of lymphoid tissue located both diffusely and in nodules spread throughout the body. It forms the mucosal immune system (**MALT / mucosa-associated lymphoid tissue**), which contains the same amount of lymphocytes as the rest of the body. It is so extensive because the mucosa of the body forms a very large surface area through which pathogenic particles can enter the body. MALT (diffuse and organised) forms **immune barriers** against pathogens in the **digestive and respiratory systems, the skin and other organs**.

1 Diffuse mucosa-associated lymphoid tissue (D-MALT)

- diffuse cells located mainly in the digestive system

2 Organised mucosa-associated lymphoid tissue (O-MALT)

- concentrations of lymphoid tissue

2.1 Gut associated lymphoid tissue (GALT)

- 2.1.1 Solitary lymphoid nodules (*nodi lymphoidei solitarii*)
- 2.1.2 Aggregated lymphoid nodules (*nodi lymphoidei aggregati*)
 - Peyer's patches (*agmina Peyerii*) in the ileum
 - Lymph nodules of vermiform appendix (*appendix vermiformis caeci*)

• 2.2 Nose associated lymphoid tissue (NALT)

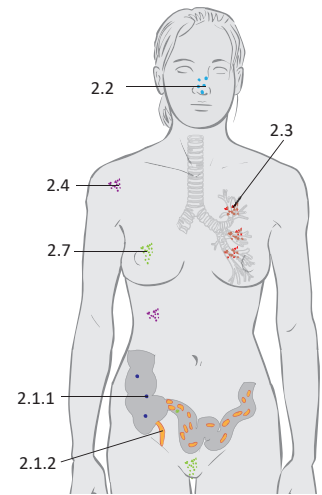
• 2.3 Bronchus associated lymphoid tissue (BALT)

• 2.4 Surface associated lymphoid tissue (SALT) – keratinocytes, mast cells, Langerhans cells

2.5 Conjunctiva associated lymphoid tissue (CALT)

2.6 Urinary tract associated lymphoid tissue (UTALT) – ureter, urinary bladder

• 2.7 Other organs: vagina, mammary gland



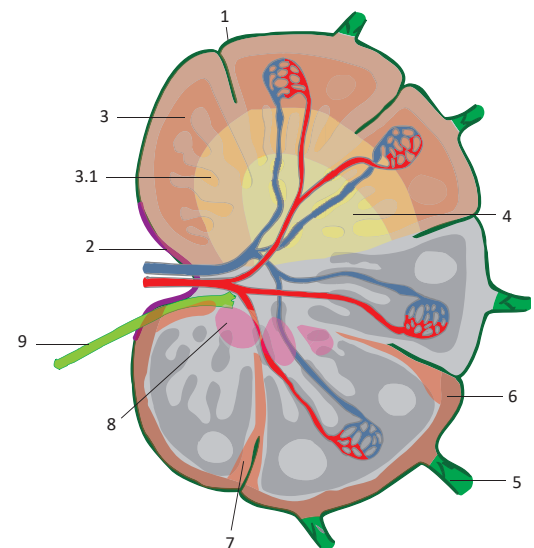
Lymph nodes are small oval-shaped organs that range in size from 1 to 25 mm. The total number of lymph nodes in the body is about 500. Lymph nodes contain **lymphocytes, macrophages and dendritic cells**. They act as filters of lymph and sites of **initiation of specific immune responses**. Naïve lymphocytes circulate between lymph nodes searching for foreign antigens on the surface of antigen presenting cells. Antigen presentation induces the activation, proliferation and differentiation of naïve lymphocytes into **mature T-lymphocytes (T-cells)** and **mature B-lymphocytes (effector B-cells / plasmocytes)**, which are capable of amounting a proper immune response.

Structure

- 1 Capsule (*capsula*)
 - 1.1 Trabeculae – extend from the internal surface of the capsule to the inner part of the lymph node
- 2 Hilum – a concave site containing the entering blood vessels and usually one exiting efferent lymphatic vessel
- 3 Cortex – contains lymph nodules, predominantly of B-cells
 - 3.1 Paracortex – contains most of the T-cells
- 4 Medulla – contains B-cells, macrophages and plasma cells

Circulation of lymph inside the lymph node

- 5 Afferent lymphatic vessels (*vasa lymphatica afferentia*)
- 6 Subcapsular/marginal sinus (*sinus subcapsularis/marginalis*)
 - is located under the capsula, and drains lymph from the afferent vessels
- 7 Internodular sinuses (*sinus internodulares*)
 - radiate from the capsule into the medulla
- 8 Medullary sinuses (*sinus lymphatici medullares*) – collecting lymph from the medulla and draining into the hilum
- 9 Efferent lymphatic vessel (*vas lymphaticum efferens*)



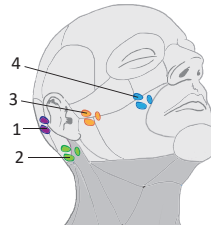
Body location

1. **Tributary region** – a specific area drained by a group of lymph nodes
2. **Regional lymph nodes** – a group of lymph nodes draining a specific anatomic region
3. **Sentinel lymph node** – the first lymph node or group of lymph nodes draining a particular organ
 - the first lymph node (or group of lymph nodes) to be invaded by metastasising cancer cells
 - swelling always indicates a pathological process

The **lymph nodes of the head** constitute a small group of only a few lymph nodes that are mostly **located around the auricles and on back of the head**.

Lymph nodes of the head

- 1 **Occipital nodes** – (*nodi lymphoidei occipitales*)
- 2 **Mastoid nodes** – (*nodi lymphoidei mastoidei*)
- 3 **Parotid nodes** – (*nodi lymphoidei parotidei*)
– divided into superficial and deep nodes
- 4 **Facial nodes** – (*nodi lymphoidei faciales*)



Only **small and inconstant lymph nodes** are located within the facial part of the head, mainly in the nasolabial groove or on the surface of the buccal fat pad.

The **tonsillar node / node of Wood** (*nodus tonsillaris*) a submandibular node located near the wall of the pharynx. Lymph from the tonsils is drained here. A painful lymphadenopathy can thus occur in tonsillitis.

The **retropharyngeal node / node of Rouvier** (*nodus retropharyngeus*) is a retropharyngeal node located in the retropharyngeal space, draining lymph from the middle ear. It can enlarge during inflammation of the middle ear.

The **jugulodigastric node / node of Küttner** (*nodus jugulodigastricus*) is a deep lateral cervical node located in the carotid triangle, at the intersection of the anterior belly of the digastric with the internal jugular vein. It drains lymph from the root of the tongue.

The **jugulo-omohyoid node** is a deep lateral cervical node located in the carotid triangle, at the intersection of the superior belly of the omohyoid with the internal jugular vein. It drains lymph from the anterior third of the tongue.

Clinical notes

The **occipital nodes** are enlarged in rubeolla/German measles and in chronic toxoplasmosis. Palpation of the lymph nodes of the head and neck is a standard part of the physical examination.

The **node of Virchow-Troiser** is located at the last segment of the thoracic duct and thus can be affected in a wide variety of tumours.

Block-neck dissection is the surgical removal of all cervical lymph nodes along with adipose tissue and/or the sternocleidomastoid muscle for treatment of head and neck malignancies.

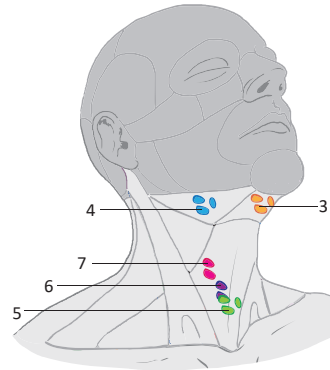
Axillary exenteration is the complete removal of the level I and II axillary nodes (according to Berg) in breast cancer. The level III lymph nodes are inaccessible due to their location behind the brachial plexus and axillary vessels.

Elephantiasis is stage three lymphoedema. Elephantiasis of the upper limb can occur as a complication of axillary exenteration in the surgical treatment of breast cancer metastasis. In such cases, it is caused by impaired lymph drainage due to the removal of lymph nodes and disruption of the deep lymphatic vessels.

The **cervical lymph nodes** are located both **superficially and deeply in the neck**. The oral cavity, pharynx and larynx drain predominantly to the deep lateral lymph nodes.

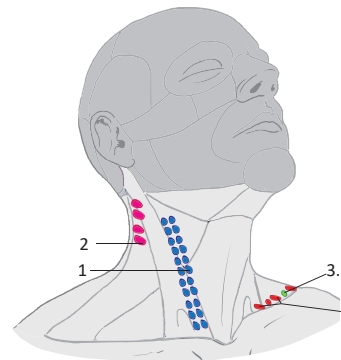
Lymph nodes classification according to the spatial arrangement

- 1 **Anterior and lateral cervical nodes**
(*nodi lymphoidei cervicales anteriores et laterales*)
- 2 **Deep and superficial cervical nodes**
(*nodi lymphoidei cervicales profundi et superficiales*)
- 3 **Submental node** (*nodi lymphoidei submentales*)
- 4 **Submandibular node** (*nodi lymphoidei submandibulares*)
- 5 **Pretracheal nodes** (*nodi lymphoidei pretracheales*)
- 6 **Paratracheal nodes** (*nodi lymphoidei paratracheales*)
- 7 **Retropharyngeal nodes** (*nodi lymphoidei retropharyngei*)
- 8 **Jugulodigastric node** (*nodus lymphoideus jugulodigastricus*)
- 9 **Jugulo-omohyoid node**
(*nodus lymphoideus juguloomohyoideus*)



Hands-on classification

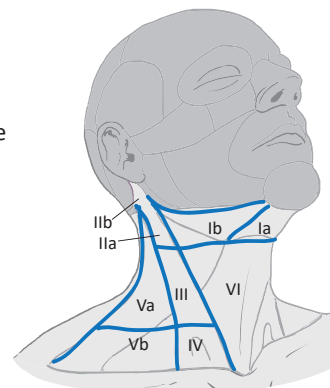
- 1 **Deep lateral cervical nodes**
(*nodi lymphoidei cervicales laterales profundi*)
– a chain of lymph nodes along the internal jugular vein
- 2 **Accessori nodes** (*nodi lymphoidei accessorii*)
– a chain of lymph nodes along the accessory nerve
- 3 **Supraclavicular nodes** (*nodi lymphoidei supraclaviculares*)
 - 3.1 **Node of Virchow-Troiser (a signal node)**
– is located above the left clavicle
– its enlargement may indicate a tumour of the stomach (70 %), left mammary gland, left lung, large intestine, left side of the neck or a primary lymphoma



Clinical classification

6 areas are distinguished for performing block-neck dissections.

- Ia** drains the lower lip, mouth, anterior third of the tongue and lower jaw
- Ib** drains the mouth, nasal cavity, skin and subcutaneous tissue of half of the face
- II a+b** drains the nasal cavity, mouth, pharynx, larynx and parotid glands
- III** drains the mouth, pharynx and larynx
- IV** drains the laryngopharynx (hypopharynx), oropharynx and larynx
- V a+b** drains the nasopharynx and oropharynx
- VI** drains the thyroid gland



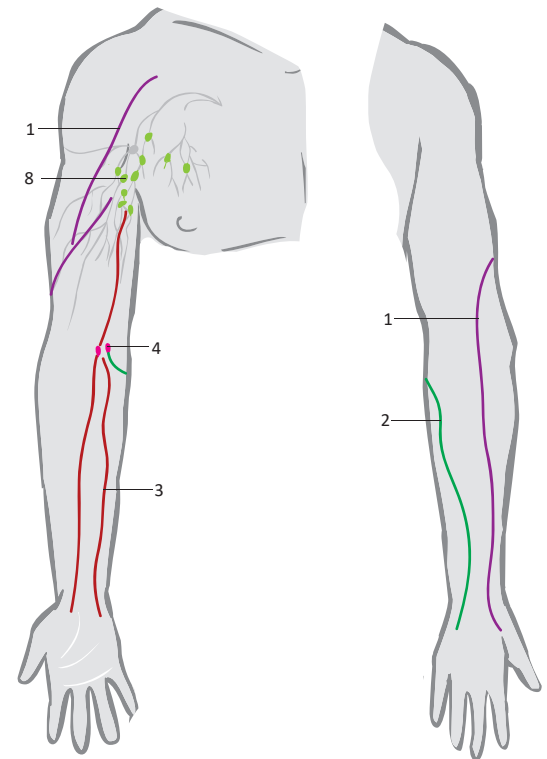
The lymph nodes of the upper limb are located mainly in **the axillary fossa** where they form a dense network that drains lymph from a large region that include **the upper limb, the mammary gland and the lateral and ventral aspects of the thorax.**

Collecting lymphatic vessels of the upper limbs (collectors)

- 1 **Lateral collectors** – into the axillary lymph nodes along the cephalic vein
- 2 **Medial collectors** – into the axillary lymph nodes along the basilic vein (with the superficial cubital lymph nodes inserted in the cubital region)
- 3 **Anterior collectors** – into the superficial cubital lymph nodes along the median antebrachial vein

Lymph nodes of the upper limb

- 4 **Superficial and deep cubital nodes**
(*nodi lymphoidei cubitales superficiales et profundi*)
- 5 **Brachial nodes** (*nodi lymphoidei brachiales*)
- 6 **Deltopectoral/Infraclavicular nodes**
(*nodi lymphoidei deltopectoriales/infraclaviculares*)
- 7 **Interpectoral nodes of Rotter**
(*nodi lymphoidei interpectoriales*)
- 8 **Axillary nodes** (*nodi lymphoidei axillares*)
– a group of up to 40 lymph nodes



Axillary lymph nodes

Anatomical classification (5 groups):

- 1 **Apical nodes** (*nodi apicales*)
- 2 **Central nodes** (*nodi centrales*)
- 3 **Humeral/lateral nodes** (*nodi humerales*)
- 4 **Subscapular/posterior nodes** (*nodi subscapulares*)
- 5 **Pectoral/anterior nodes** (*nodi pectorales*)
 - 5.1 **Sorgius node**
– located at the 2nd or 3rd slip of the serratus anterior
– the sentinel lymph node of the mammary gland

Clinical classification (according to Berg)

- **Pectoralis minor** (*musculus pectoralis minor*)

I. Under pectoralis minor

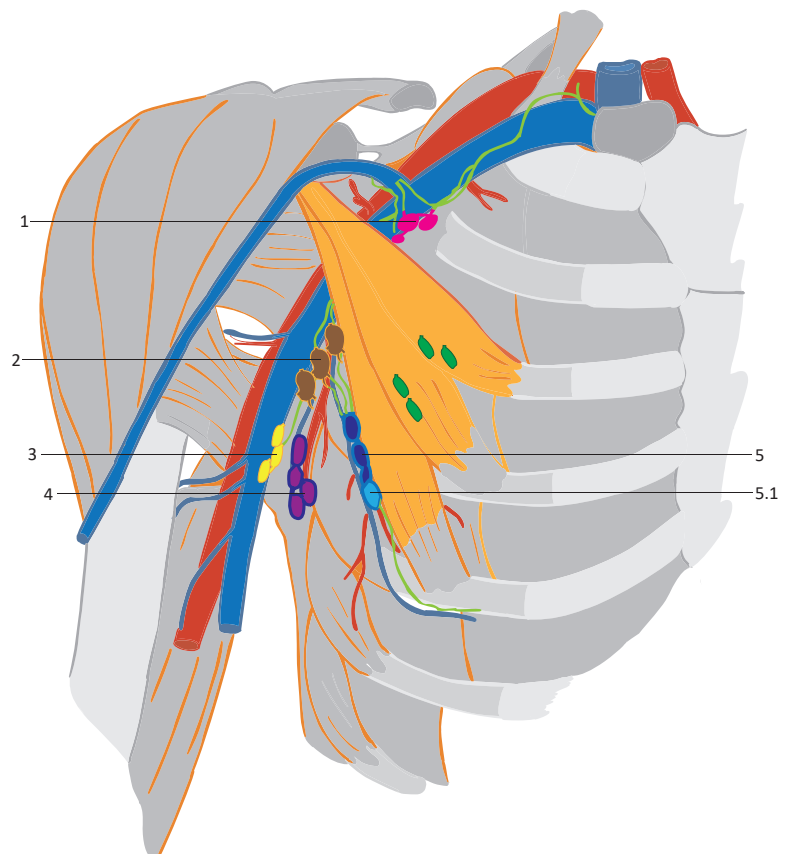
- Pectoral nodes – ventrally
- Subscapular nodes – dorsally
- Humeral nodes – lateral

II. Behind pectoralis minor

- Central nodes
- Apical nodes – partially
- Interpectoral nodes

III. Behind the neurovascular bundle

- Apical nodes



The lymph nodes of the thorax are located in the mediastinum. They drain **the intrathoracic organs and the mammary gland**. Unfortunately this location is poorly accessible, which makes it difficult to differentiate pathological enlargement of these lymph nodes.

Classification according to the position in the mediastinum

1 Superior

- 1.1 Superior tracheobronchial lymph nodes
(*nodi lymphoidei tracheobronchiales superiores*)

- 1.2 Paratracheal lymph nodes
(*nodi lymphoidei paratracheales*)

2 Inferior anterior (*nodi lymphoidei mediastinales anteriores*)

- 2.1 Paramammary lymph nodes
(*nodi lymphoidei paramammarii*)

- 2.2 Parasternal lymph nodes
(*nodi lymphoidei parasternales*)

- 2.3 Intercostal lymph nodes
(*nodi lymphoidei intercostales*)

- 2.4 Superior diaphragmatic lymph nodes
(*nodi lymphoidei phrenici superiores*)

- 2.5 Prepericardial lymph nodes
(*nodi lymphoidei prepericardiaci*)

3 Inferior middle

- 3.1 Bronchopulmonary lymph nodes
(*nodi lymphoidei bronchopulmonales*)

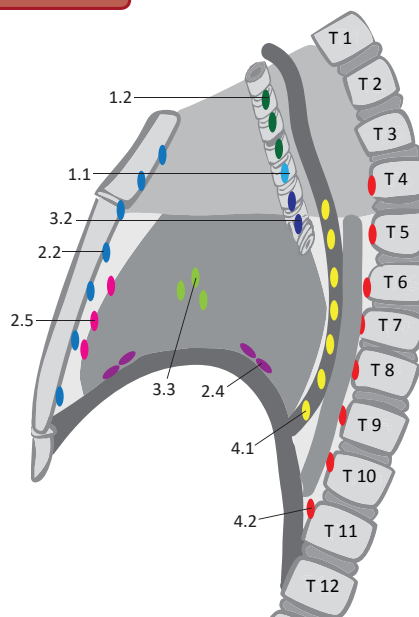
- 3.2 Inferior tracheobronchial lymph nodes
(*nodi lymphoidei tracheobronchiales inferiores*)

- 3.3 Lateral pericardial lymph nodes
(*nodi lymphoidei pericardiaci laterales*)

4 Inferior posterior (*nodi lymphoidei mediastinales posteriores*)

- 4.1 Juxta-oesophageal lymph nodes (*nodi lymphoidei juxtaoesophageales*)

- 4.2 Prevertebral lymph nodes (*nodi lymphoidei prevertebrales*)



Scheme of the thoracic lymph nodes

Classification according to the organs drained

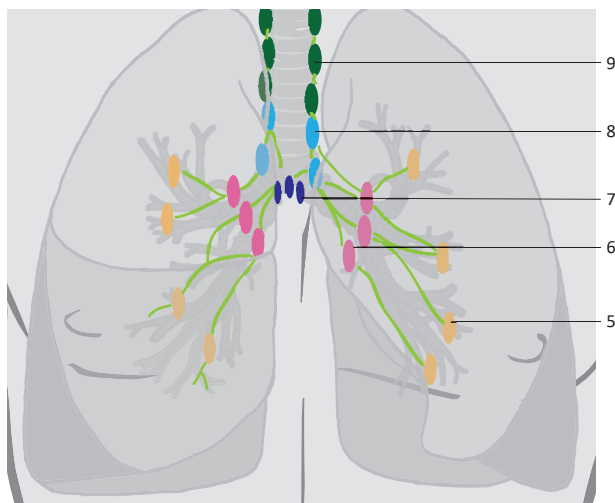
- 5 Intrapulmonary lymph nodes (*nodi lymphoidei intrapulmonales*)
– are located along the course of the bronchi inside the lung parenchyma

- 6 Bronchopulmonary lymph nodes (*nodi lymphoidei bronchopulmonales*)
– are located in the hilum of the lung, close to the entry of the main bronchi into the lungs

- 7 Inferior tracheobronchial lymph nodes (*nodi lymphoidei tracheobronchiales inferiores*)
– are located below the bifurcation of the trachea

- 8 Right and left superior tracheobronchial lymph nodes
(*nodi lymphoidei tracheobronchiales superiores dextri et sinistri*)
– are located lateral to the bifurcation of the trachea

- 9 Paratracheal lymph nodes (*nodi lymphoidei paratracheales*) – are located along the trachea



Lymph from the thoracic wall (from the parasternal lymph nodes) is drained on the left side to the thoracic duct, while on the right side to the right lymphatic trunk. Lymph from the other groups of lymph nodes is drained to the mediastinal and lumbar lymph nodes.

The **paraaortic, aortocaval and paracaval lymph nodes** are clinical terms for the lumbar lymph nodes.

The **intestine** can be considered to be the largest lymphatic organ due to the huge amount of lymphatic nodules within its wall.

Clinical notes

Hodgkin's lymphoma is a cancer of the lymphatic system that usually manifests in the lymph nodes. In the majority of cases, it occurs in the lymph nodes of the neck, axilla or mediastinum. It spreads to adjacent lymph nodes or to the systemic circulation. It typically occurs in young adults and the elderly.

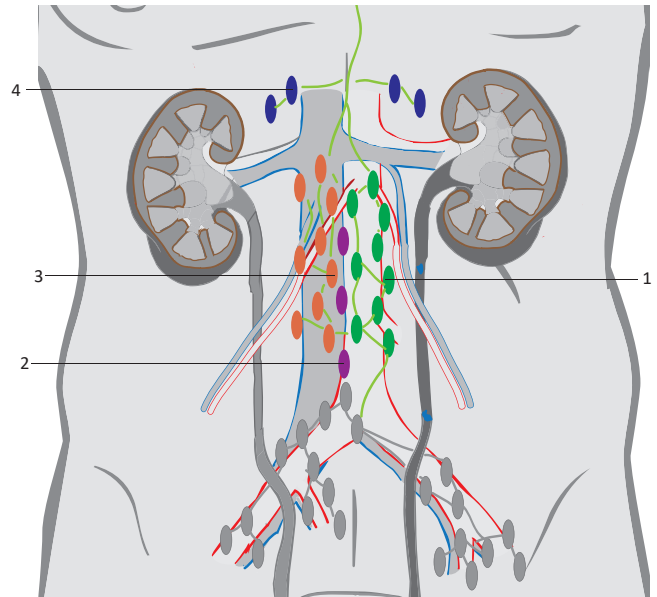
Paraaortic lymphadenectomy is the surgical removal of the lumbar lymph nodes in malignancies of the ovaries, testes, uterus or kidneys.

Retroperitoneal lymphadenopathy refers to an enlargement of the lumbar lymph nodes. It can be caused by the spread of tumour cells. It manifests by various symptoms, such as swelling of the lower limbs due to obstruction of lymphatic drainage and compression of the large veins, renal failure due to compression of the ureters and pain caused by compression of nerves.

The lymph nodes of the intraperitoneal organs drain to the coeliac lymph nodes and then via two or three intestinal trunks to the thoracic duct. The lumbar lymph nodes collect lymph from the pelvic region, lower extremities, abdominal wall, paired organs of the abdominal cavity, gonads and the organs of the inframesocolic part of the peritoneal cavity. Lymph from the lumbar lymph nodes is subsequently drained to the paired lumbar trunks and to the thoracic duct. Lymph from the unpaired organs of the abdominal cavity drains firstly to the lymph nodes situated along the vessels within the walls of these organs and then to the lymph nodes located along the unpaired branches of the abdominal aorta. Lymph from the coeliac lymph nodes is collected by the intestinal trunks and drained into the thoracic duct. Lymph from the superior mesenteric lymph nodes is collected by the coeliac and lumbar lymph nodes. Lymph from the inferior mesenteric lymph nodes is drained solely to the lumbar lymph nodes.

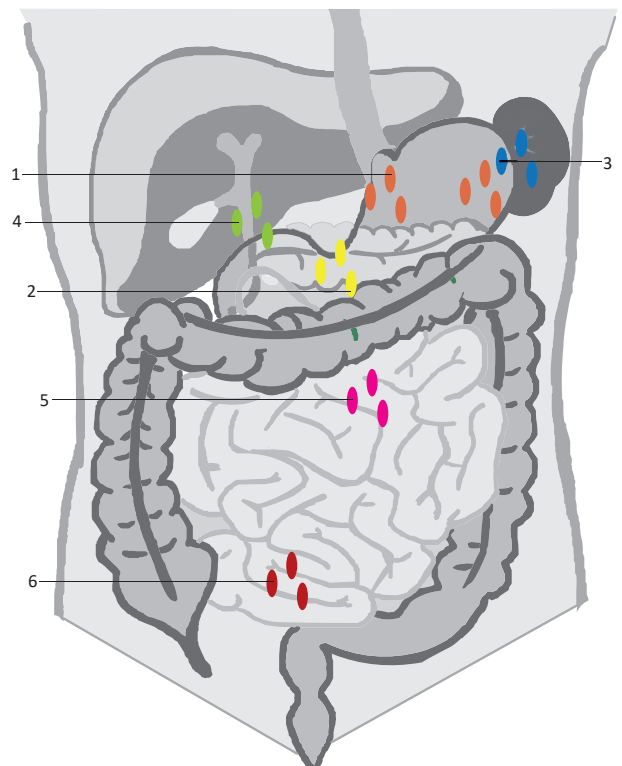
Parietal lymph nodes (nodi lymphoidei parietales)

- 1 Left lumbar lymph nodes
(*nodi lymphoidei lumbales sinistri*)
 - 1.1 Lateral aortic, preaortic, postaortic lymph nodes
(*nodi lymphoidei aortici laterales, preaortici, retroaortici*)
- 2 Intermediate lumbar lymph nodes
(*nodi lymphoidei lumbales intermedii*)
- 3 Right lumbar lymph nodes
(*nodi lymphoidei lumbales dextri*)
 - 3.1 Lateral caval, precaval, retrocaval lymph nodes
(*nodi lymphoidei cavales laterales, precavales, retrocavales*)
- 4 Inferior diaphragmatic lymph nodes
(*nodi lymphoidei phrenici inferiores*)
- 5 Inferior epigastric lymph nodes
(*nodi lymphoidei epigastrici inferiores*)



Visceral lymph nodes (nodi lymphoidei viscerales)

- 1 Gastric, gastroepiploic, pyloric and pancreaticoduodenal lymph nodes
(*nodi lymphoidei gastrici, gastroepiploici, pylorici et pancreaticoduodenales*)
- 2 Pancreatic lymph nodes (*nodi lymphoidei pancreatici*)
- 3 Splenic lymph nodes (*nodi lymphoidei splenici*)
- 4 Hepatic lymph nodes (*nodi lymphoidei hepatici*)
- 5 Superior mesenteric lymph nodes
(*nodi lymphoidei mesenterici superiores*)
 - 5.1 Juxta-intestinal mesenteric lymph nodes
(*nodi lymphoidei juxtaintestinales*)
 - 5.2 Central superior mesenteric lymph nodes
(*nodi lymphoidei mesenterici superiores centrales*)
 - 5.3 Ileocolic lymph nodes (*nodi lymphoidei ileocolici*)
 - 5.4 Precaecal and retrocaecal lymph nodes
(*nodi lymphoidei precaecales et retrocaecales*)
 - 5.5 Appendicular lymph nodes (*nodi lymphoidei appendiculares*)
 - 5.6 Mesocolic and paracolic lymph nodes
(*nodi lymphoidei mesocolici et paracolici*)
 - 5.7 Right, middle and left colic lymph nodes
(*nodi lymphoidei colici dextri, medii et sinistri*)
- 6 Inferior mesenteric lymph nodes
(*nodi lymphoidei mesenterici inferiores*)
 - 6.1 Sigmoid lymph nodes (*nodi lymphoidei sigmoidei*)
 - 6.2 Superior rectal lymph nodes (*nodi lymphoidei rectales superiores*)



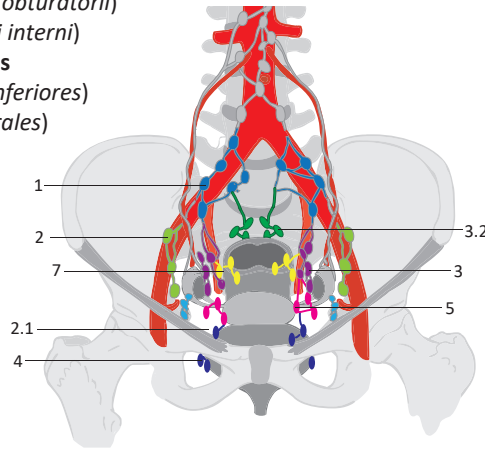
Lymph from the pelvic organs drains into **the internal and external iliac lymph nodes**. **Lymph from the gonads drains into the lumbar lymph nodes** as these organs originated in the superior lumbar region.

Parietal lymph nodes (nodi lymphoidei parietales)

- 1 **Common iliac lymph nodes** (*nodi lymphoidei iliaci communes*)
- 2 **External iliac lymph nodes** (*nodi lymphoidei iliaci externi*)
 - 2.1 **Obturator lymph nodes** (*nodi lymphoidei obturatorii*)
- 3 **Internal iliac lymph nodes** (*nodi lymphoidei iliaci interni*)
 - 3.1 **Superior and inferior gluteal lymph nodes** (*nodi lymphoidei gluteales superiores et inferiores*)
 - 3.2 **Sacral lymph nodes** (*nodi lymphoidei sacrales*)

Nodi lymphoidei viscerales

- 4 **Paravesical lymph nodes** (*nodi lymphoidei paravesicales*)
- 5 **Parauterine lymph nodes** (*nodi lymphoidei parauterini*)
- 6 **Paravaginal lymph nodes** (*nodi lymphoidei paravaginales*)
- 7 **Pararectal lymph nodes** (*nodi lymphoidei pararectales*)



Plague buboes are swollen lymph nodes in the inguinal region. They are associated with the bubonic plague caused by the bacteria *Yersinia pestis*.

The external iliac lymph nodes form three bands along the homonymous veins. These lymph nodes collect lymph from the lower limbs, walls of the lesser pelvis, inferior part of the abdominal wall, parts of the uterus and urinary bladder and external genital organs.

The internal iliac lymph nodes collect lymph from the walls and organs of the lesser pelvis, the deep tissue of the perineum, external genital organs and gluteal region.

The superficial inguinal lymph nodes are positioned within the subcutaneous layer of the femoral triangle, close to the saphenous hiatus. Lymph is drained through the cribriform fascia to the deep inguinal lymph nodes. The deep inguinal lymph nodes are the tributary lymph nodes of the lower limbs (except for the gluteal region), they are located at the bottom of the femoral triangle (within the iliopectineal fossa) close to the vascular space.

Lymphatic drainage of the heart: the left atrium and ventricle are drained via the left lymphatic outflow system to the inferior tracheobronchial nodes and then into the right lymphatic duct. Lymph from the right atrium and ventricle is drained via the right lymphatic outflow system into the anterior mediastinal lymph nodes and then into the thoracic duct.

8.7 Lymph nodes of the lower limb – Nodi lymphoidei membri inferioris

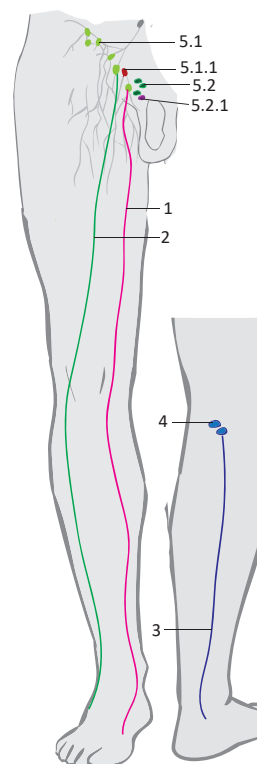
The lymph nodes of the lower limbs are **mainly located in the inguinal region and femoral triangle**. They **drain lymph from the lower extremities, external genital organs and part of the anterior and lateral abdominal walls**.

Collecting lymphatic vessels of the lower limbs (collectors)

- 1 **Medial collecting vessels** (10–15) – are located along the great saphenous vein and drain into the superficial inguinal lymph nodes
- 2 **Lateral collecting vessels** (1–3) – are located along the tributaries of the great saphenous vein and drain into the superficial inguinal lymph nodes
- 3 **Posterior collecting vessels** (2–3) – are located along the small saphenous vein and drain into the superficial popliteal lymph nodes

Lymph nodes

- 4 **Popliteal lymph nodes** (*nodi lymphoidei poplitei*) – are divided into a superficial and a deep group
- 5 **Inguinal lymph nodes** (*nodi lymphoidei inguinales*)
 - 5.1 **Superficial inguinal lymph nodes** (*nodi lymphoidei inguinales superficiales*) 8–12
 - **superomedial, superolateral and inferior group**
 - drain almost the entire lower limb (except for its posterior surface), the skin and subcutaneous tissue of the gluteal region, perineum, anus, penis/clitoris, vaginal vestibule, scrotum/labia majora, horns of the uterus, and inferior anterior abdominal wall
 - 5.1.1 **Lymph node of Cabanas** (*nodus lymphoideus inguinalis superficialis superomedialis*) – the sentinel lymph nodes of the penis
 - 5.2 **Deep inguinal lymph nodes** (*nodi lymphoidei inguinales profundus*) 2–5
 - 5.2.1 **Node of Cloquet–Rosenmüller** (*nodus lymphoideus inguinalis profundus proximalis*) – is situated medially in the vascular space (*lacuna vasorum*)



Clinical notes

Pelvic lymphadenectomy is the surgical removal of the lymph nodes in the pelvic cavity. It is usually performed in malignancies and their lymphatic metastases in this region.

The obturator lymph nodes may be affected in uterine malignancies. They are located along the obturator vessels.

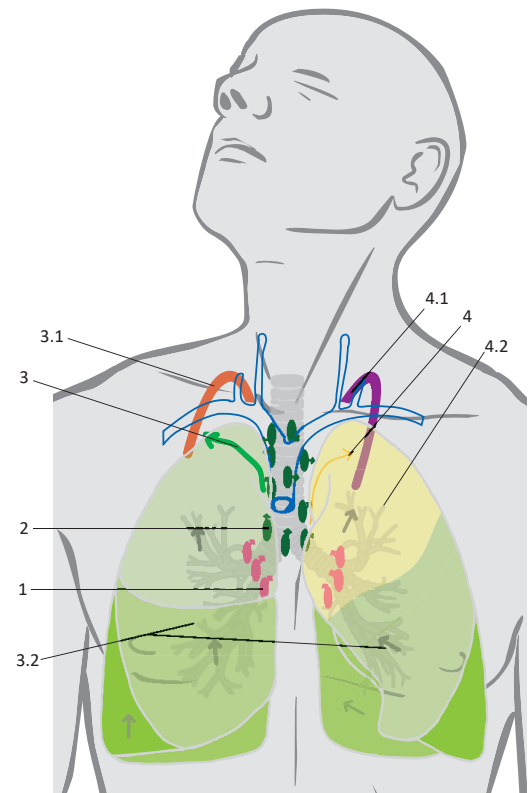
The sacral lymph nodes may be affected in malignancies of the cervix of the uterus, prostate and caecum.

Respiratory system

- 1 **Nose** – submandibular lymph nodes
- 2 **Nasal cavity** – retropharyngeal and deep cervical lymph nodes
- 3 **Larynx** – deep cervical and paratracheal lymph nodes
- 4 **Trachea** – paratracheal and superior, right inferior and left inferior tracheobronchial lymph nodes
- 5 **Bronchi** – intrapulmonary lymph nodes
 - into the bronchopulmonary and subsequently to the tracheobronchial lymph nodes

Lungs

- 1 **Bronchopulmonary lymph nodes**
- 2 **Inferior, right and left tracheobronchial lymph nodes**
- 3 **Right bronchomediastinal trunk**
 - 3.1 **Into the right lymphatic duct**
 - 3.2 **Drainage area:** whole right lung, inferior lobe of the left lung, both lingual segments of the superior lobe of the left lung (IV, V)
- 4 **Left bronchomediastinal trunk**
 - 4.1 **Into the thoracic duct**
 - 4.2 **Drainage area:** superior lobe of the lung (I–III) except for the both lingual segments



Digestive system

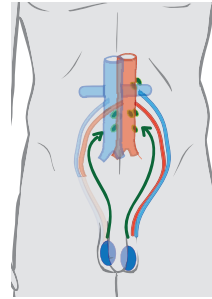
- 1 **Oral cavity**
 - 1.1 **Face** – facial lymph nodes (on the surface of the buccinator), submandibular lymph nodes
 - 1.2 **Lips** – submandibular and submental lymph nodes
 - 1.3 **Gingiva** – submandibular lymph nodes
 - 1.4 **Teeth**: submandibular and deep cervical lymph nodes
 - 1.5 **Tongue**
 - 1.5.1 **Apex** – submental lymph nodes
 - 1.5.2 **Margin** – submandibular lymph nodes
 - 1.5.3 **Radix and corpus** – deep cervical lymph nodes and also to the contralateral lymph nodes due to an incomplete lingual septum
 - 1.6 **Palatine tonsil** – node of Wood at the level of the angle of the mandible and deep cervical lymph nodes
 - 1.7 **Salivary glands**
 - 1.7.1 **Submandibular gland** – submandibular lymph nodes
 - 1.7.2 **Sublingual gland** – submandibular lymph nodes
 - 1.7.3 **Parotid gland** – parotid lymph nodes, superficial and deep cervical lymph nodes
- 2 **Pharynx** – retropharyngeal lymph nodes
- 3 **Oesophagus**
 - 3.1 **Cervical part** – deep cervical lymph nodes
 - 3.2 **Thoracic part** – tracheobronchial and posterior mediastinal lymph nodes
 - 3.3 **Abdominal part** – left gastric lymph nodes
- 4 **Stomach** – coeliac lymph nodes and subsequently to the thoracic duct and into the left supraclavicular lymph nodes (lymph nodes of Virchow-Troisier which are palpable)
 - 4.1 **Body of stomach** – right and left gastric lymph nodes, right and left gastroepiploic lymph nodes
 - 4.2 **Fundus of stomach** – splenic lymph nodes and left gastroepiploic lymph nodes
 - 4.3 **Pylorus** – pyloric lymph nodes
- 5 **Duodenum** – pyloric, hepatic and mesenteric lymph nodes
 - to the coeliac lymph nodes and then to the thoracic duct
- 6 **Jejunum and ileum** – superior mesenteric lymph nodes,
 - to the coeliac and lumbar lymph nodes and then to the thoracic duct
- 7 **Large intestine**
 - 7.1 **Terminal part of the ileum, caecum, vermiform appendix and adjacent part of the colon**
 - into the ileocolic, prececal, retrocecal, appendicular, paracolic lymph nodes
 - 7.2 **Ascending colon** – right colic lymph nodes
 - 7.3 **Right two-thirds of the transverse colon** – middle colic lymph nodes
 - 7.4 **Left one-third of the transverse colon, descending colon and sigmoid colon** – left colic lymph nodes
 - 7.5 **Rectum** – inferior mesenteric lymph nodes, internal and external iliac lymph nodes, sacral lymph nodes
- 8 **Liver** – hepatic and coeliac lymph nodes and other via the diaphragm
 - (to the anterior and posterior mediastinal lymph nodes, superior and inferior, parasternal lymph nodes)
- 9 **Pancreas** – hepatic, coeliac, pancreatic and splenic lymph nodes

Urinary tract

- 1 **Kidneys** – drain into the lumbar lymph nodes
 - three complex lymphatic networks from the parenchyma and the fibrous and adipose capsule
- 2 **Urinary bladder** – drains into the paravesical lymph nodes, then into the common, internal and external iliac lymph nodes
- 3 **Female urethra** – drains into the paravesical lymph nodes, then into the common, internal and external iliac lymph nodes

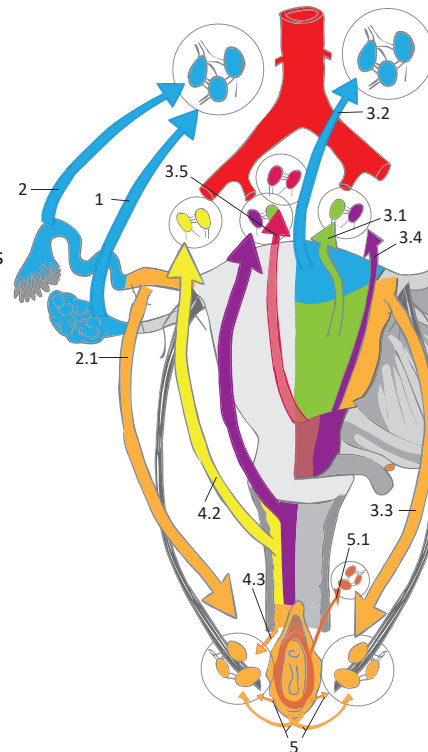
Male genital system

- 1 **Testes** – drain into the lumbar lymph nodes
- 2 **Epididymis** – drains into the lumbar lymph nodes
- 3 **Seminal vesicles** – drain into the internal and external iliac lymph nodes
- 4 **Prostate** – drains into the internal and external iliac lymph nodes, sacral lymph nodes
- 5 **Scrotum** – drains into the superficial inguinal lymph nodes
- 6 **Penis** – the skin of the penis drains into the superficial inguinal lymph nodes
 - the node of Cabanas is the sentinel node of the penis
- 6.1 **Glans** – drains into the deep inguinal lymph nodes and the internal iliac lymph nodes
- 6.2 **Corpora cavernosa and male urethra** – drain into the external iliac lymph nodes



Female genital system

- 1 **Ovaries** – drain into the lumbar lymph nodes
- 2 **Uterine tubes** – drain into the lumbar lymph nodes
 - 2.1 **Isthmus** – drains into the superficial inguinal lymph nodes
- 3 **Uterus**
 - 3.1 **Corpus** – drains into the internal iliac lymph nodes
 - 3.2 **Fundus** – drains into the lumbar lymph nodes
 - 3.3 **Margins** – drain into the superficial inguinal lymph nodes alongside the round ligament of the uterus
 - 3.4 **Anterior part of the cervix** – drains into the internal iliac lymph nodes
 - 3.5 **Posterior part of the cervix** – drains into the sacral lymph nodes
- 4 **Vagina**
 - 4.1 **Upper part cranial to the hymen** – drains into the internal iliac lymph nodes
 - 4.2 **Lower part cranial to the hymen** – drains into the external iliac lymph nodes
 - 4.3 **Caudally to the hymen** – drains into the external iliac lymph nodes
- 5 **External female genital organs**
 - 5.1 **Clitoris and labia minora** – drain into the deep inguinal lymph nodes

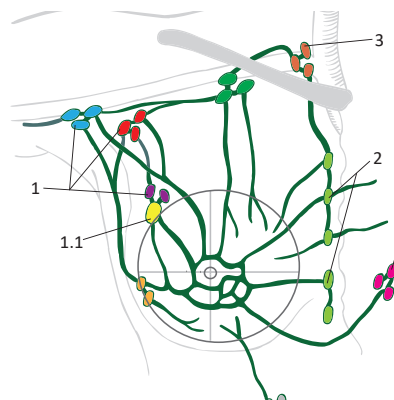


Lymphatic drainage of the female genital organs

Mammary gland

The lymphatic drainage of the mammary gland is complex. It begins at the subareolar plexus of Sappey.

- 1 **Axillary lymph nodes**
 - 1.1 **Node of Sargius**
 - the sentinel node of the mammary gland
 - the caudal-most lymph node of the pectoral lymph nodes
 - located at the 2nd or 3rd slip of the anterior serratus muscle
- 2 **Parasternal lymph nodes**
- 3 **Supraclavicular lymph nodes**
- 4 **Axillary lymph nodes**

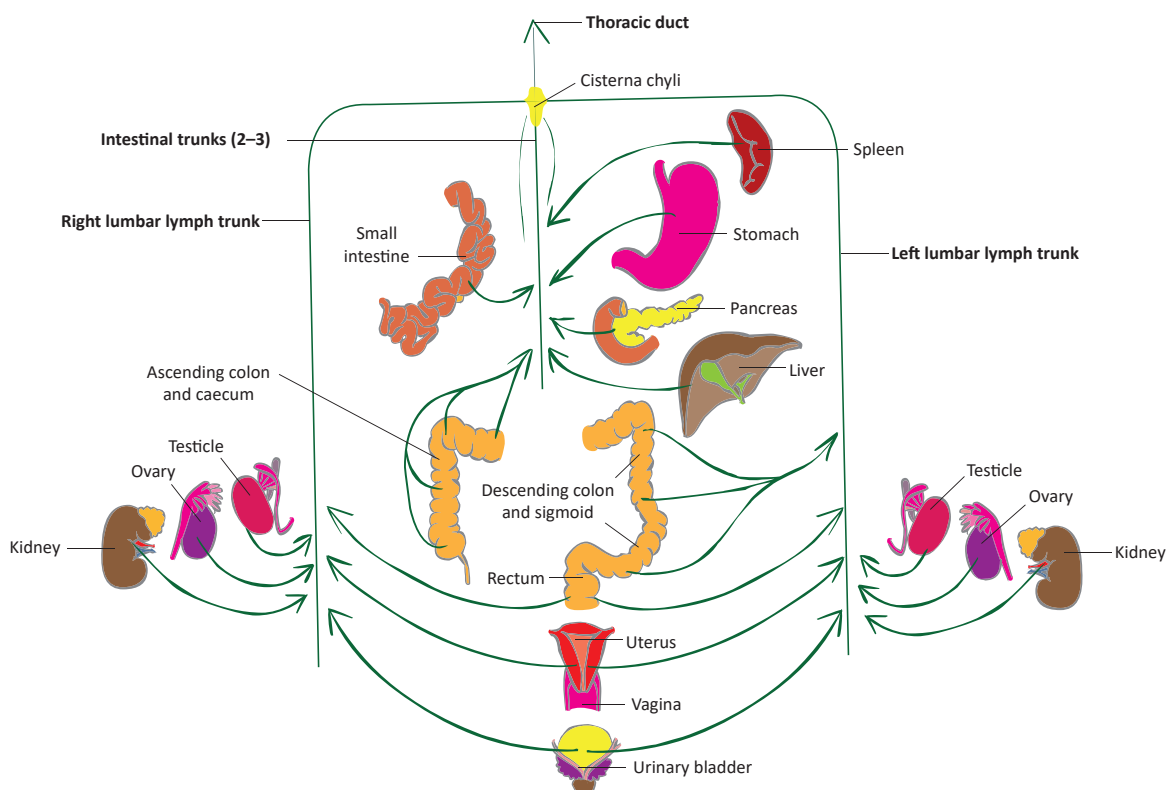
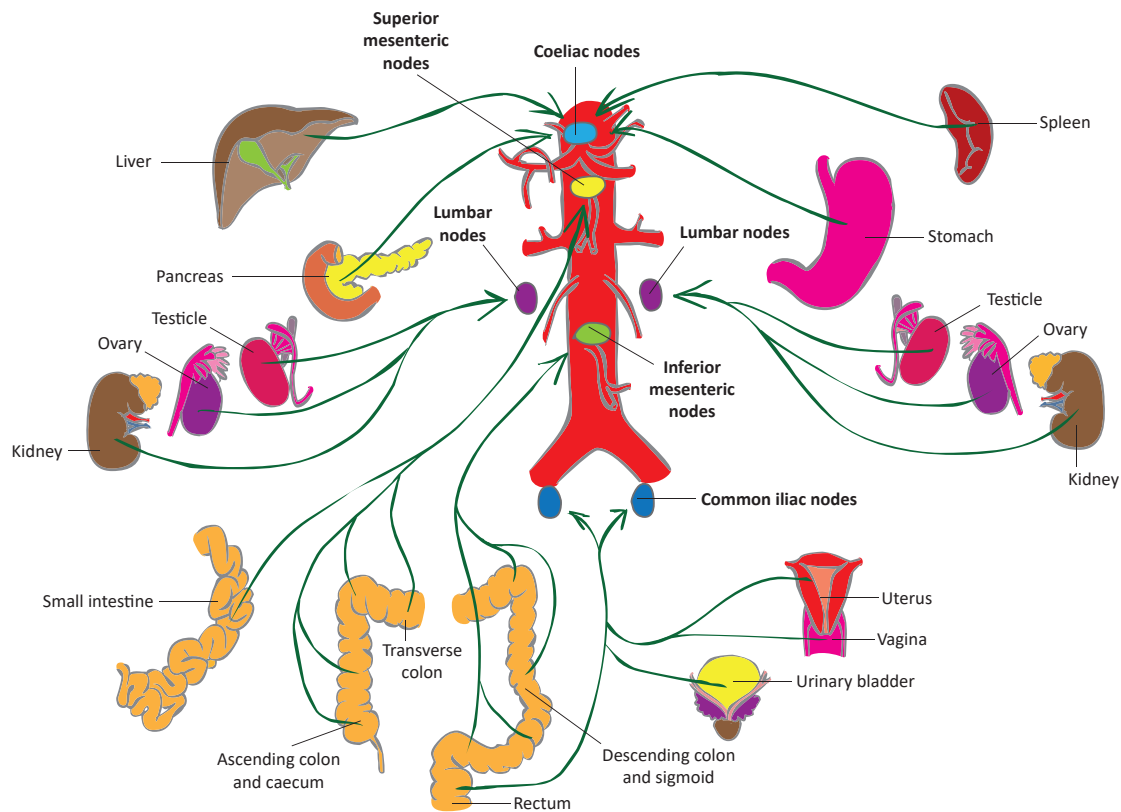


Lymphatic drainage of the mammary gland

Clinical notes

The majority of mammary tumours occur in the superior lateral quadrant of the mammary gland. Therefore metastases usually spread to the axillary lymph nodes.

The lymphatic systems of both mammary glands are interconnected. This explains why metastases may spread to the contralateral mammary gland or even (rarely) to the contralateral axillary lymph nodes.



I. Lymphatic and immune system

1. Explain the abbreviation "nn.l.". (p. 318)
2. State the 6 organs of the immune and lymphatic systems and divide them into the primary and secondary lymphatic organs. (p. 318)
3. State the functions of the lymphatic system. (p. 318)
4. Explain in which organs the specific immune reaction is activated. (p. 318)
5. State the five parts of the human body, where lymphatic vessels are not present. (p. 318)

II. Lymphatic vessels

6. Explain the differences between the lymphatic and vascular capillaries. (p. 319)
7. State the collectors of the upper and lower limbs. (p. 319)
8. Describe the parts of the body, which are drained into the thoracic duct. (p. 319)
9. State the three trunks from which the thoracic duct is created and state the location of the thoracic duct (p. 319)

III. Thymus

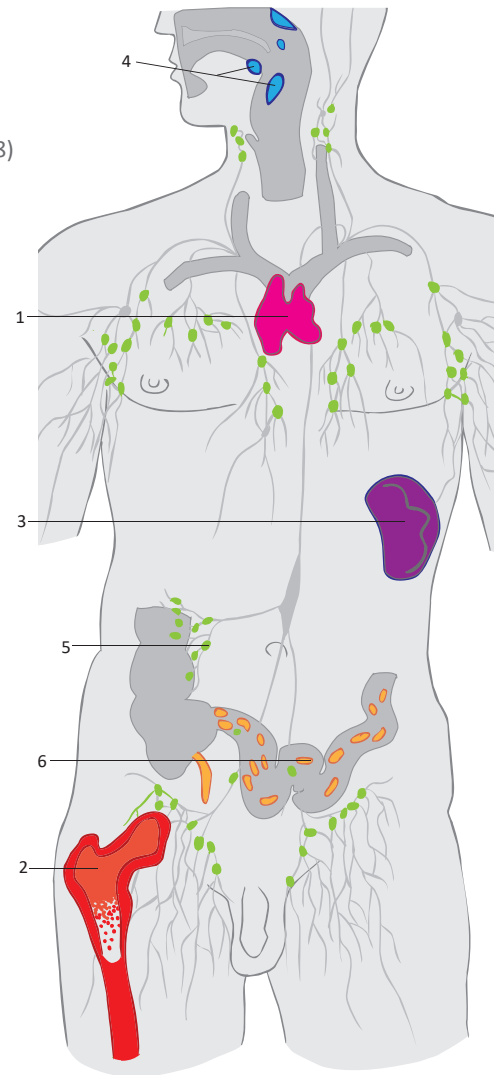
10. Describe the topography of the thymus. (p. 320)
11. State the main functions of the thymus. (p. 320)
12. Explain the term involution of the thymus. (p. 320)

IV. Bone marrow

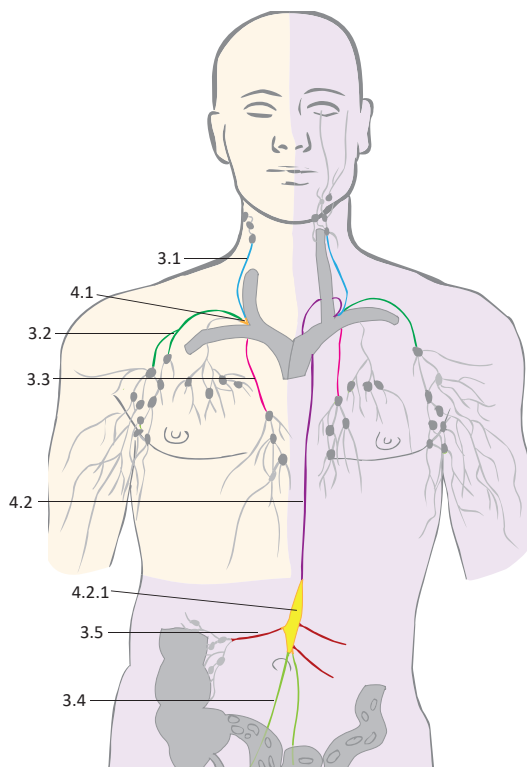
13. Describe the changes of location of the bone marrow. (p. 320)
14. Explain the conversion of red marrow into yellow marrow. (p. 320)
15. Explain whether it is possible for yellow marrow to change back into the red marrow. (p. 320)

V. Spleen

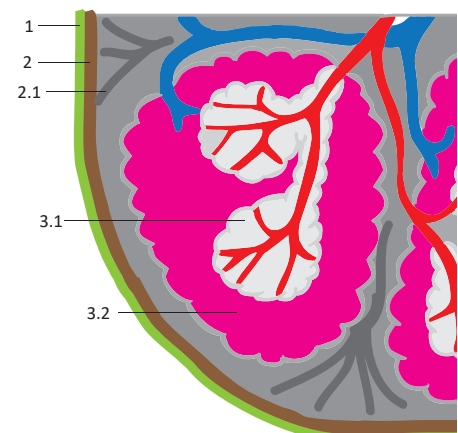
16. Explain the main functions of the spleen. (p. 321)
17. Describe the syntopy and topography of the spleen. (p. 321)
18. State the structure where vessels enter the spleen. (p. 321)
19. Name the part of the spleen where T-lymphocytes are located. (p. 321)
20. Describe the structure and main functions of the red pulp. (p. 321)
21. Explain the differences between the open and closed circulation inside the spleen. (p. 321)



Describe the organs of the immune and lymphatic system



Describe the lymphatic vessels



Describe the internal structures of the spleen

VI. Tonsils

22. State 4 tonsils of the lymphatic ring of Waldeyer. (p. 322)
23. List the paired and unpaired tonsils and give their locations. (p. 322)
24. Describe the location of the pharyngeal tonsil. (p. 322)

VII. Non-encapsulated lymphoid tissue

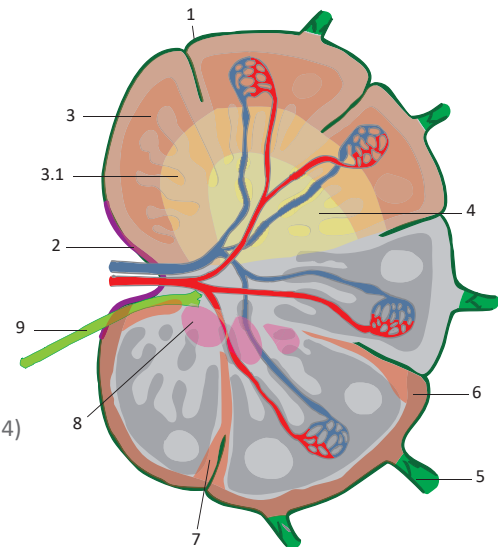
25. State several tissues/structures where the non-capsulated lymphatic tissue is located. (p. 323)
26. Explain the abbreviation GALT. (p. 323)
27. Explain the term Peyer's patches and describe their structure and function. (p. 323)

VIII. Lymph nodes

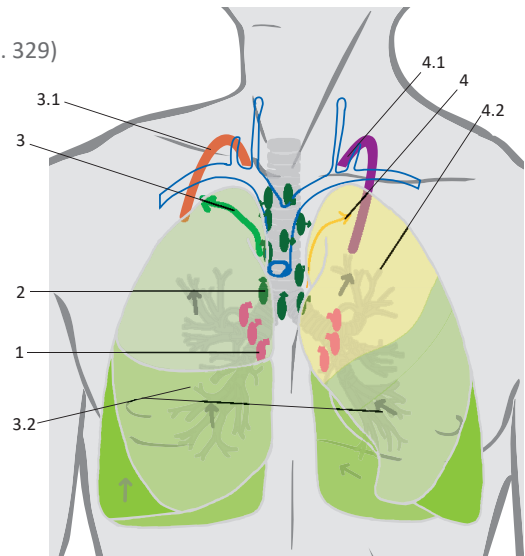
28. State the parts of the lymph node containing B-lymphocytes. (p. 323)
29. Explain the term sentinel lymph node and give some examples. (p. 323)
30. Explain where the lymph node of Rouvière can be palpated in children suffering from infections of the middle ear. (p. 324)
31. State the location of the node of Virchow-Troisier and the areas/organs it drains. (p. 324)
32. State the organs that in the presence of a malignancy may be associated with an enlargement of the node of Virchow-Troisier. (p. 324)
33. Describe the clinical classification of neck the lymph nodes (six areas of cervical block dissection in otorhinolaryngology). (p. 324)
34. State the location of the node of Sörorius and the region and organ/s it drains. (p. 325)
35. Describe the topography of the muscle which divides the axillary lymph nodes into three clinically relevant groups. (p. 325)
36. State the location and tributary area of the node of Cabanas. (p. 328)

VIII. Tributary lymph nodes

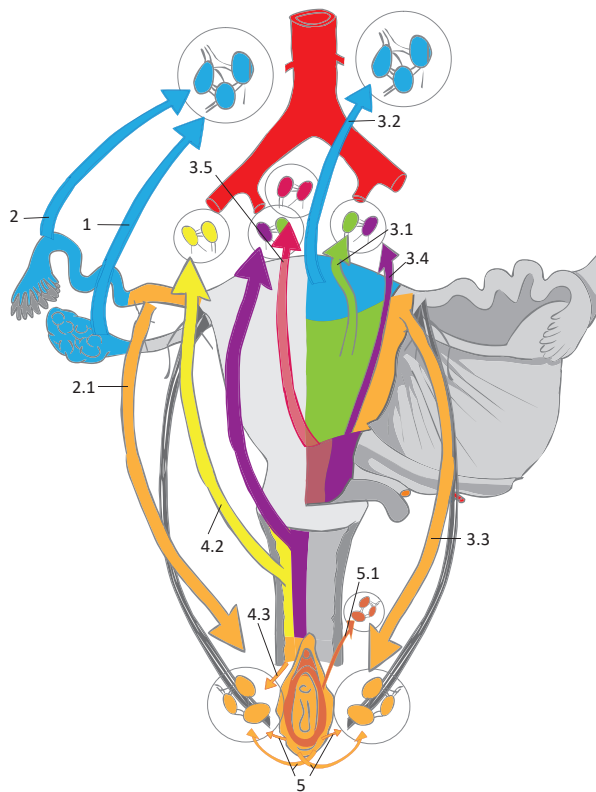
37. State the lymph nodes collecting lymph from the superior lobe of the left lung. (p. 329)
38. State lymph nodes collecting lymph from the testis. (p. 330)
39. Describe and compare the lymphatic drainage of the internal and external female genital organs. (p. 330)
40. Describe the drainage of the mammary gland in great detail. (p. 330)



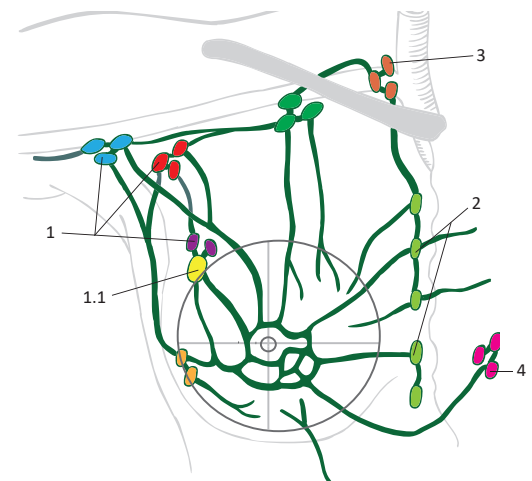
Describe the structure of the lymph node



Describe the lymphatic drainage of the lungs



Describe the lymphatic drainage of the female genital organs



Describe the lymphatic drainage of the mammary gland

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1. AGUR, A. M. R. and A. F. DALLEY. Grant's Atlas of Anatomy. 13th Ed. Philadelphia: Lippincot Williams & Wilkins, 2012, 888 p. ISBN 978-1-60831-756-1.
2. BRAUNWALD, E. and R. O. BONOW. Braunwald's heart disease: a textbook of cardiovascular medicine. 9th Edition. Philadelphia: Saunders/Elsevier, 2012. 2136 p. ISBN 978-0-8089-2436-4.
3. CHUNG, K. W.; CHUNG, H. M. and N. L. HALLIDAY. BRS Gross Anatomy. 8th Ed. Philadelphia: Lippincott Williams & Wilkins, 2015. 544 p. ISBN 978-1-4511-9307-7.
4. LANZ, T. and W. WACHSMUTH. Praktische Anatomie. 7 bands. Berlin: Springer, 2003, 3658 p. ISBN 978-3-540-40571-9.
5. MILLS, S. E. Histology for pathologists. 4th Ed. Philadelphia: Lippincot Williams & Wilkins, 2012. 1328 p. ISBN-13: 978-1-4511-1303-7.
6. NETTER, F. H. Atlas of Human Anatomy. Professional Edition. 6th Ed. Oxford: Elsevier. 2014. 640 p. ISBN 978-1-455-75888-3.
7. SNELL, R. S. Clinical anatomy by regions. 9th Ed. Philadelphia: Lippincott Williams & Wilkins, 2012. 768 p. ISBN 978-1-60913-446-4.
8. WILLIAMS, P. L., ed. Gray's Anatomy: the anatomical basis of medicine and surgery. 38th Ed. New York: Churchill Livingstone, 1995, 2092 p. ISBN 0-443-05717-6.